A Brief Derive Version History

Martin Hepperle - 2022-2026

1980

1994

1987

The following pages present a tabulation of versions of Derive and related products like muLISP, muSIMP and, muMath.

The LISP language implementation muLISP was the foundation of a whole family of products. As LISP was rather uncommon for the average user, muSIMP was a simplified user and programming interface wrapped around muLISP. User friendliness was further increased by implementing muMATH on top of muSIMP. Based on the positive experience with muMATH and muSIMP, Derive was finally created with an ever more capable kernel and a more sophisticated user interface. Overall efficiency was improved by implementing it directly in muLISP.

Unfortunately, most printed manuals for these software products seem to be unavailable in scanned form, which is a pity. For example, scans of the following manuals would be very welcome to preserve this piece of software history:

- “muLISP-77 Reference Manual”
- “muLISP-79 Reference Manual”
- “muLISP-81 Reference Manual”
- “muLISP-82 Reference Manual”
- “muLISP-85 Reference Manual”, The Soft Warehouse, 137 p., 1986
- “muLISP-86 Reference Manual”, The Soft Warehouse, 408 p., 1986
- muMath-83 for MS-DOS systems

It is unclear, whether a version muLISP-84 ever existed.

Albert D. Rich, the main author of muLISP, died at age 74 on 11 August 2023. Together with David Stoutemyer he founded Soft Warehouse in 1979, which was active until 1999. He was the main author of muLISP. Using muLISP, Albert and David developed muMATH, followed by Derive. Later, Theresa Shelby adapted Derive to the Microsoft Windows environment, greatly enhancing its user interface and graphics capabilities. Derive was still based on muLISP, but the Windows version of muLISP was not sold as a product.

The first versions of muLISP, muSIMP and muMath ran on 8-bit CP/M systems and were later ported and further developed under 16-bit MS-DOS and MS-Windows. Derive started its life on the MS-DOS platform. The development of Derive finally ended up at Texas Instruments. Here, it was the starting point for computer algebra systems that were implemented in many of TI's graphing pocket calculators.

The following table highlights the growth of the muLISP system (MS-DOS version, if not noted otherwise, without extensions loaded) by counting the available primitives. Additional functions were included through external files, as needed.

Table 1: Number of LISP primitives available in the various muLISP versions.

Product	Primitives
muLISP-80 (CP/M)	89, 8080 version before Microsoft licensing
muLISP-80 2.0 (CP/M)	88, version licensed to Microsoft
muLISP-83 4.11 (CP/M)	118, version licensed to Microsoft
muLISP-83 4.11	121
muLISP-85 5.01	303
muLISP-86 5.10	354
muLISP-87 6.00	411, irrational and transcendental functions: IRRATNAL.LSP
muLISP-87 6.10	415, irrational and transcendental functions: IRRATNAL.LSP
muLISP-90 7.20	421, irrational and transcendental functions: IRRATNAL.LSP

Table 2: Licensed distribution of Soft Warehouse products by Microsoft between 1980 and 1986.

Product	**Platform**	**1980**	**1981**	**1982**	**1983**	**1984**	**1985**	**1986**
muLISP	Apple II (Z80)			■	■	■		
	Apple II (6502)			■				
	CP/M-80	■	■	■	■	■		
	TRS-80			■	■			
	MS-DOS				■	■	■	■
muMATH/muSIMP								
	Apple II (Z80)	■	■	■	■	■		
	Apple II (6502)	■	■	■				
	CP/M-80	■	■	■	■			
	TRS-80			■	■			
	MS-DOS				■	■	■	■

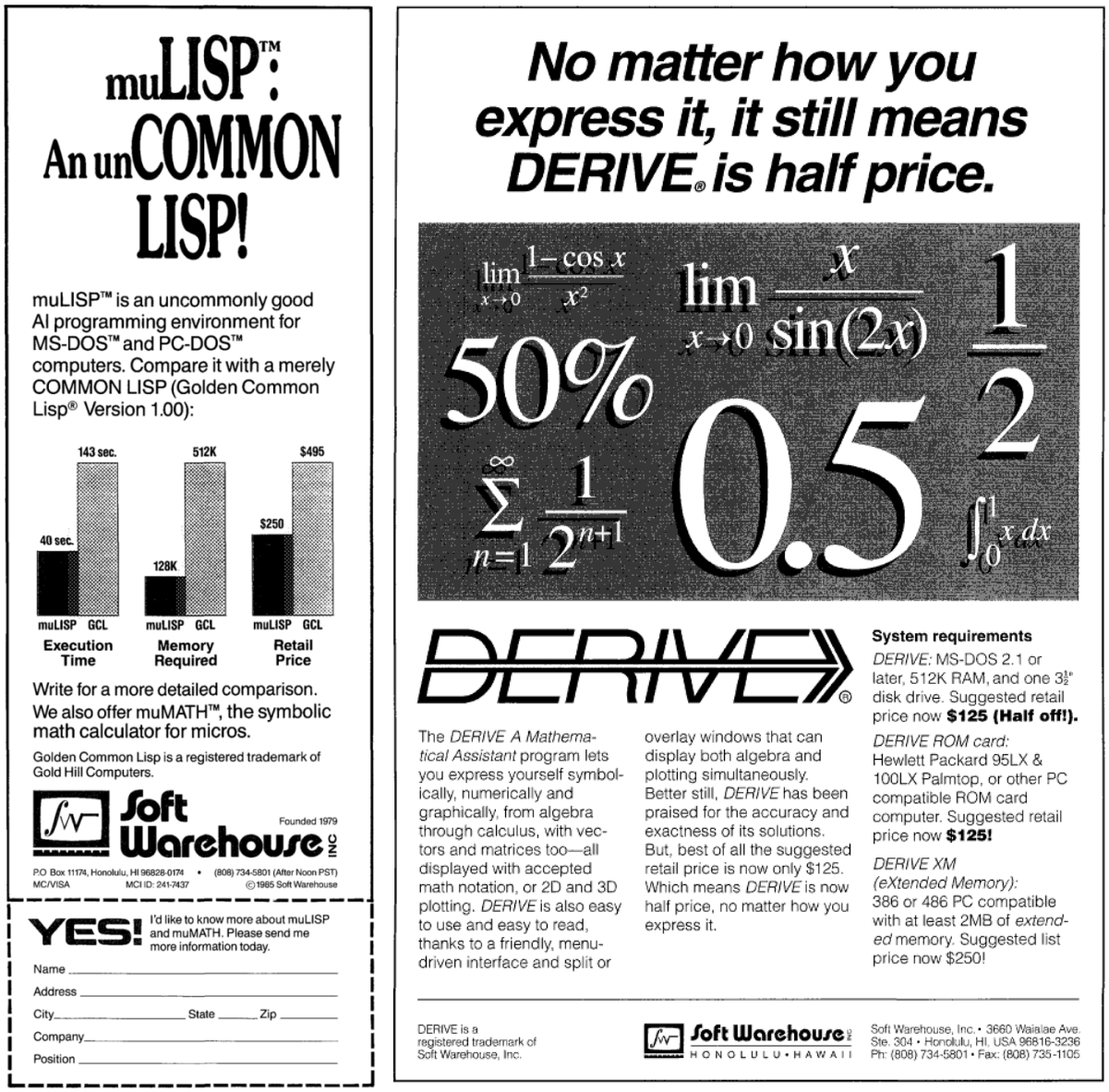


Figure 1: Advertisements for muLISP and muMATH in 1985 and Derive in 1994.

Figure 2: Derive was also available as a PCMCIA ROM card for the HP 95LX Palmtop (Screen shows the built-in Lotus 1-2-3).

DOS Extender (XM) Versions

These versions became available with muLISP-90 and require an 80386 CPU to access Extended Memory. The following numbers were obtained in DOS-Box.

muLISP-90 (RECLAIM) returns 340'467 bytes free
muLISP-90 XM (RECLAIM) returns 15'002'227 bytes free

Year	Product	Operating System and Comments
1977	muLISP-77	"micro LISP", implemented on IMSAI 8080
1977	muSIMP-77	"microcomputer Structured Implementation Language", written in muLISP-77
1979	January 1	*The Soft Warehouse* founded by Albert D. Rich, David R. Stoutemyer
1979	muMath-79	CP/M (8080, Z80), TRS-80 DOS, written in muSIMP-77
1980	muLISP-80	CP/M, with muStar, (8080), 10/06/80, before licensing to Microsoft
1980	muMath-80	Apple II (ADIOS, 6502), Apple II (CP/M , Z80 card), TRSMATH-80 1.0
1980	muLISP-80	CP/M, Version 2.0, licensed by Microsoft
1980	muLISP-80	CP/M, Version 2.03, licensed by Microsoft
1980	muMath-80	CP/M, written in muSIMP-80 2.02 (Microsoft)
1980	muMath-80	CP/M, written in muSIMP 2.03, licensed by Microsoft
1981	muMath-80	CP/M, written in muSIMP 2.10, 04/25/81, licensed by Microsoft
1981	muMath-80	CP/M, written in muSIMP 2.12, 07/09/81, licensed by Microsoft
1981	muMath-80	CP/M, Osborne 1, written in muSIMP-80 2.14, 12/19/81 (Microsoft)
1981	muLISP-80	CP/M, Version 2.15, licensed by Microsoft
1981	muLISP-81	IBM PC and CP/M
1982	muLISP-82	IBM PC and CP/M, see: PC Tech Journal April 1984; Micro/Systems Journal, May/June, 1985
1982	muMath-82	IBM PC, see PC-Magazine Review, December, 1983
1982	muMath-80	Apple II/ADIOS-81, 01/29/82, written in muSIMP 2.15, 03/01/82 (Microsoft)
1983	muLISP-83	CP/M-80, Soft Warehouse Version 4.10 03/07/84, licensed by Microsoft
1984	muLISP-83	CP/M-80, Soft Warehouse Version 4.11 03/22/84, licensed by Microsoft
1984	muLISP-83	IBM PC, Soft Warehouse Version 4.11 03/22/84, licensed by Microsoft
1985	muMath-83	IBM PC MS-DOS, Version 4.12 (03/30/85), licensed by Microsoft, last version of muMATH
1985		Company name changed to *Soft Warehouse Hawaii*.
1985	muLISP-85	IBM PC MS-DOS, Microsoft LISP Version 5.01 09/15/85
1986	muLISP-86	IBM PC MS-DOS, Microsoft LISP Version 5.10 01/06/86
1986	muLISP-86	IBM PC MS-DOS, Microsoft LISP Version 5.11 03/11/86
1987	muLISP-87	IBM PC MS-DOS, Soft Warehouse Version 6.01 06/17/87
1988	muLISP-87	IBM PC MS-DOS, Soft Warehouse Version 6.03, 07/12/88
1988	muLISP-87	IBM PC, MS-DOS, Soft Warehouse Version 6.10, 12/07/88 (RECLAIM: 340'592 bytes)
1988	Derive 1.00	MS-DOS, implemented in muLISP, released on October
1988	Derive 1.02	MS-DOS, written in muLISP-87 (© 1983, 1985, 1986, 1987)
1988	Derive 1.13	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1988	Derive 1.20	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1988	Derive 1.51	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1988	Derive 1.53	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1988	Derive 1.56	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1988	Derive 1.60	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1988	Derive 1.61	MS-DOS
1988	Derive 1.62	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1989	Derive 2.00	MS-DOS, price $200

Year	Product	Operating System and Comments
1990	muLISP-90	IBM PC, Version 7.10, $150, up to 640 KB
1990	muLISP-90 XM	IBM PC, Version 7.10, $300, up to 16 MB with PharLap Extender
1990	Derive 2.013	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1990	Derive 2.023	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1990	Derive 2.033	MS-DOS, written in muLISP-87 (© 1983, 1987, 1989)
1990	Derive 2.05	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1990	Derive 2.053	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1990	Derive 2.06	MS-DOS, written in muLISP-90, advertised by EduCALC for HP-95 palmtop
1990	Derive 2.073	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1990	Derive 2.083	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1991		Derive User Group (DUG) founded, newsletter published up to 2024
1992	Derive 2.50	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990), HP 95LX Card
1992	Derive 2.51	MS-DOS, written in muLISP-90
1992	Derive 2.54	MS-DOS, written in muLISP-90
1992	Derive 2.55	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1992	Derive 2.55 XM	MS-DOS, written in muLISP-90 XM Version 7.16 (02/10/93), uses PharLap extender
1992	Derive 2.56	MS-DOS, written in muLISP-90
1992	Derive 2.56 XM	MS-DOS, written in muLISP-90 XM
1993	muLISP-90 XM	MS-DOS, Version 7.16 (02/10/93)
1993	Derive 2.57	MS-DOS, written in muLISP-90
1993	Derive 2.57 XM	MS-DOS, written in muLISP-90 XM Version 7.17 (07/14/93), uses PharLap extender
1993	Derive 2.58	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1993	Derive 2.60	MS-DOS, written in muLISP-90
1993	Derive 2.60 XM	MS-DOS, uses PharLap extender
1994	muLISP-90	IBM PC, Version 7.20 (02/07/94) , (RECLAIM: 360'467 bytes)
1994	muLISP-90 XM	IBM PC, Version 7.20 (02/07/94), uses PharLap extender, (RECLAIM: 15'002'227 bytes)
1994	Derive 3.00	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990), export for Acrospin for 3D
1994	Derive 3.00y	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1994	Derive 3.00y XM	MS-DOS, written in muLISP-XM 7.21, uses PharLap extender
1994	Derive 3.01	MS-DOS
1994	Derive 3.02	MS-DOS
1994	Derive 3.05	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1995	Derive 3.06 XM	MS-DOS, written in muLISP-XM 7.21 (08/09/95), uses PharLap extender
1995	muLISP-90 XM	MS-DOS, Version 7.30 (10/13/95)
1995	Derive 3.10	MS-DOS, written in muLISP-XM 7.30 (10/13/95), uses PharLap extender
1995	Derive 3.10 G	MS-DOS, written in muLISP-XM 7.30 (10/13/95), uses PharLap extender, German
1995	Derive 3.10 XM	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990), uses PharLap extender
1995	Derive 3.10 XMG	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990), uses PharLap extender, German
1995	Derive 3.11 XM	MS-DOS, uses PharLap extender
1995	Derive 3.12	MS-DOS
1995	Derive 3.13	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1995	Derive 3.14	MS-DOS, written in muLISP-90 (© 1983, 1987, 1990)
1995	Derive 3.14 XM	MS-DOS, uses PharLap extender
1996		new versions for Windows written in special version "muLISP for Derive"
1996	Derive 4.00	MS-DOS, combines 16/32-bit; Version for Windows 3 and 95 released in October
1996	Derive 4.01	MS-DOS, 16/32-bit; Windows, but not for Windows XP
1996	Derive 4.02	MS-DOS, 16/32-bit; Windows, but not for Windows XP

Year	Product	Operating System and Comments
1996	Derive 4.03	MS-DOS, 16/32-bit; Windows, but not for Windows XP
1996	Derive 4.04	MS-DOS, written in muLISP 7.40 (08/28/96), 16/32-bit; Windows
1996	Derive 4.05	MS-DOS, 16/32-bit, written in muLISP Version 7.40 (04/15/97)); Windows
1996	Derive 4.05a	MS-DOS, 16/32-bit; Windows 3.x
1996	Derive 4.06	MS-DOS, written in muLISP-XM 7.40 (06/25/97), 16/32-bit; Windows
1996	Derive 4.07	MS-DOS, 16/32-bit, written in muLISP-XM 7.40, 16/32-bit; Windows
1996	Derive 4.08 XM	MS-DOS, 16/32-bit
1996	Derive 4.09	MS-DOS, 16/32-bit; Windows (have update only)
1996	Derive 4.10	MS-DOS, 16/32-bit; Windows (have update only)
1996	Derive 4.11	MS-DOS, written in muLISP 7.40 (06/04/98); Windows 3.1 with Win32s, 95, 98, XP, NT
	Derive 4.13	MS-DOS; Windows, but not for Windows 2000
1999		takeover by Texas Instruments, MS-DOS version dropped
2000		a new branch is started by rewriting the core in "C" for TI-92 and future calculators
2000	Derive 5.00	Windows 3.x, still following the main branch with its LISP core
2000	Derive 5.01	Windows 95, 98, NT (16 June 2000)
2000	Derive 5.02	Windows 95, 98, NT (30 June 2000)
2001	Derive 5.03	Windows 95, 98, NT, 2000 (15 January 2001)
2001	Derive 5.04	Windows 95, 98, NT, 2000 (11 April 2001)
2001	Derive 5.05	Windows 95, 98, NT, 2000 (5 December 2001)
2002	Derive 5.06	Windows 95, 98, ME, NT, 2000, XP (10 October 2002), written in muLISP for Derive 7.43 (03/06/01)
2003	Derive 6.00	Windows 98, ME, 2000, XP
2004	Derive 6.01	Windows 2000, XP (not: 95, 98, ME, NT 4.0 due to Unicode), (1 March 2004)
2004	Derive 6.10	Windows 98, ME, 2000, XP (October 2004), written in muLISP for Derive 7.45 (08/17/04), connects to TI-89, TI-89 Titanium, TI-92+, TI Voyage 200
2007		Texas Instruments ends development and distribution of Derive

Notes:

- Colors used in this table:
  - red: I have archived a copy of the files.
  - black: must exist; I found references to or screenshots of these versions. An archived copy does not exist.
- The prefix "mu" stands for "micro".
- There are probably more intermediate versions not listed here, as Soft Warehouse often released quick fixes for problems reported by users. I only listed those, which I found referenced in books, journal articles and papers.

Companion Programs

The Derive versions for MS-DOS starting with Derive 3.0 supported the external program *Acrospin*. *Acrospin* (by David Parker) could display three projections of dimensional wireframe meshes and allowed dynamic rotation and translation. The program used the VGA screen and double buffering to achieve a nice, smooth display. The data was transferred from the Plot menu via an .ACD text file which contains lists of coordinates and straight line connectivities. See also Derive Newsletter #24.

Localization

Most messages from Derive versions since 2.053 can be localized by adding a language file “DERIVE.LAN”. This file contains pairs of English and foreign translations for the Prompt and Message Lines, but not for the menus. The menu bar is still in English and some of the prompts are not fully translated, for example the yes/no question prompt “Y/N” which expects a ‘Y’ or ‘N’ key to be pressed.

Starting with Derive version 3.05 the localization of the menus was possible by adding a menu file “DERIVE.MNU”. It contains a hierarchical LISP list containing the translated menu structure and the English commands to execute.

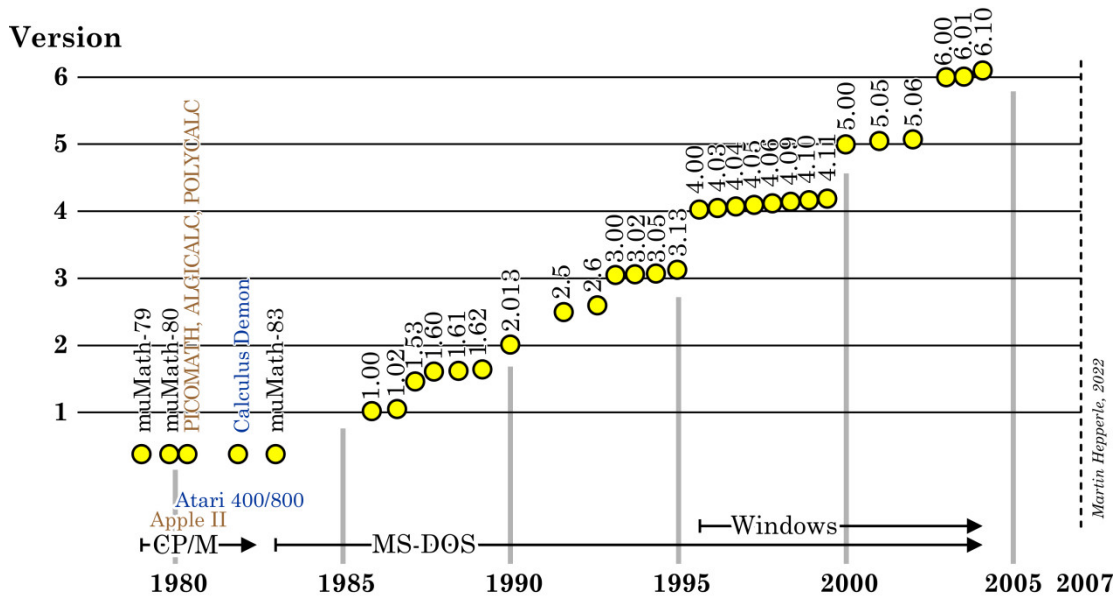


Figure 3: Derive version numbers versus publication year.

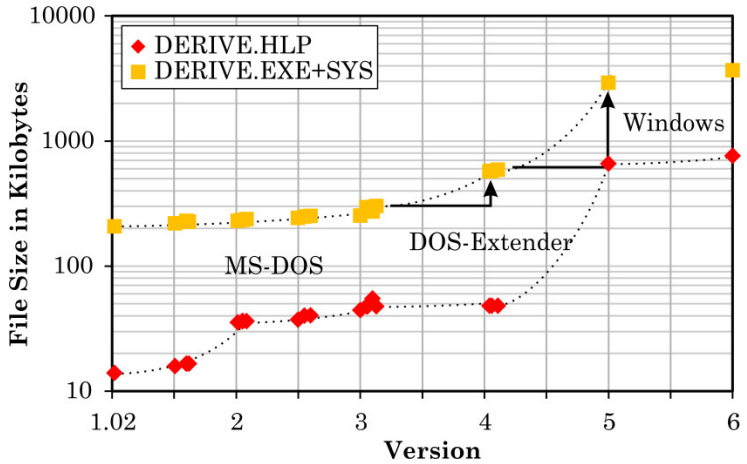
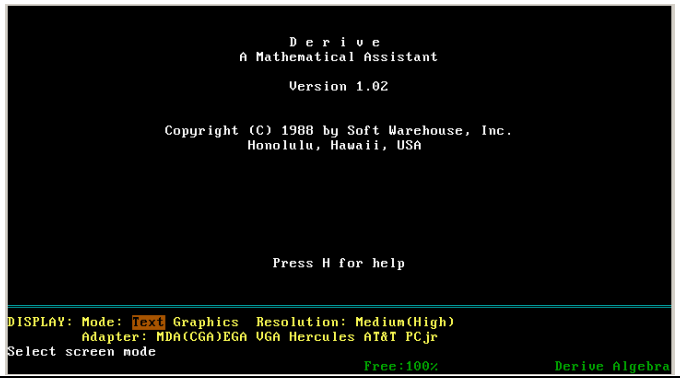
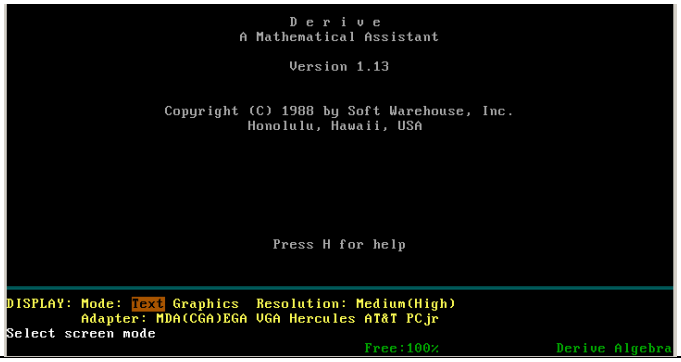
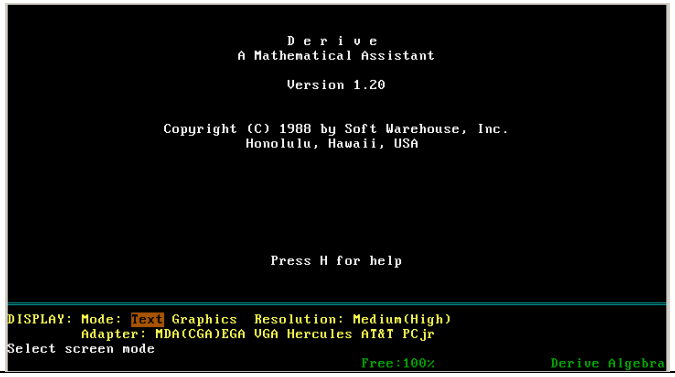
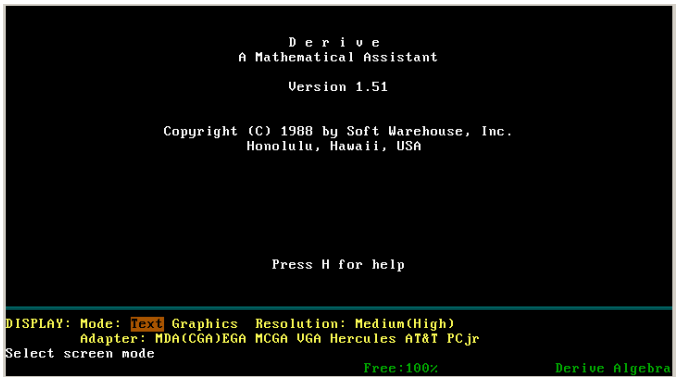
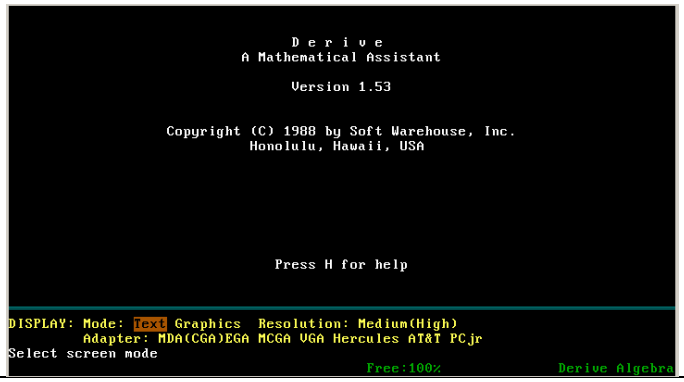
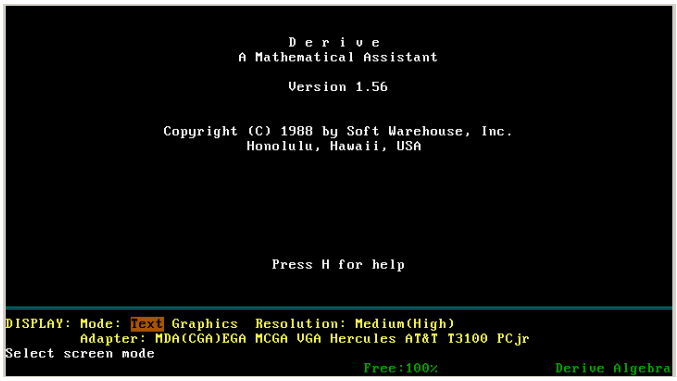


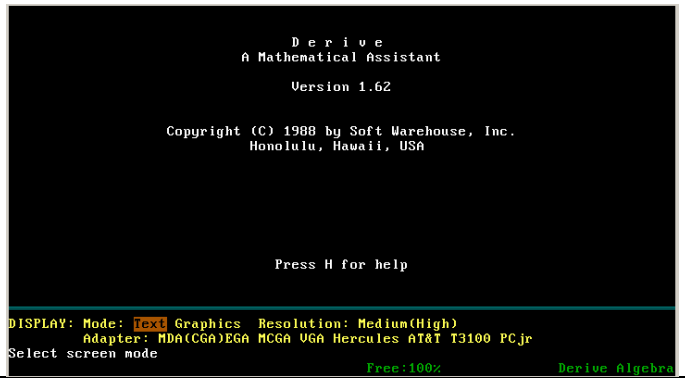
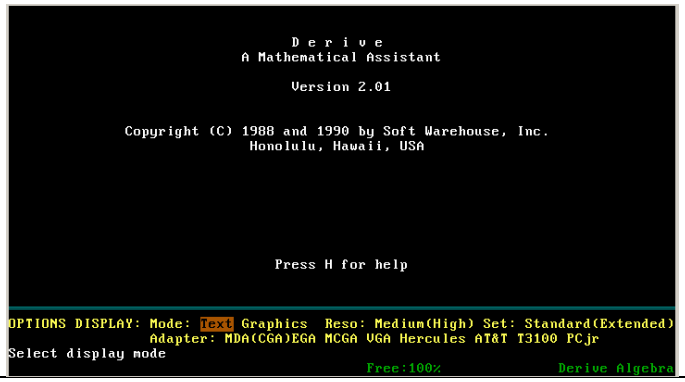
Figure 4: Growth of the main Derive files versus publication year. The size of the executable programs is related to the increasing number of features and also to changes of the underlying operating platform (moving from DOS to Windows). The size and quality of the help files also increased dramatically when they were recreated using the Windows help system.

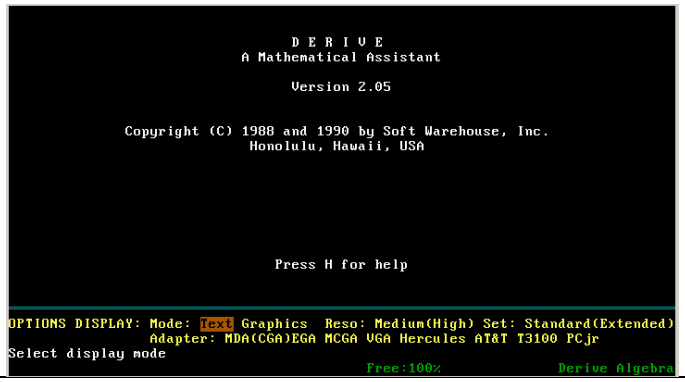
Derive Opening Screens

Derive can be installed and run on modern systems using a virtual machine with MS-DOS installed or using a simulation environment like DOSBox. The following screenshots were generated in DOSBox with Derive's Options / Display active to show the graphics cards supported by each version.

	1988 Version 1.02 This version already supports most graphics adapters of that era including VGA. The AT&T 6300 provided a resolution of 640x400.
	1988 Version 1.13
	1988 Version 1.20
	1988 Version 1.51 Support for the new IBM PS/2 Multi-Color-Graphics-Adapter has been added.

	1988 Version 1.53
	1988 Version 1.56 This version added a 640 x 400 pixel display driver for the Toshiba T3100 laptop computer (with the same screen and memory structure as the AT&T 6300/Olivetti M24 and the HP Vectra Multimode Adapter).
	1988 Version 1.60
	1988 Version 1.61

	1988 Version 1.62
	1990 Version 2.013
	1990 Version 2.023
	1990 Version 2.033

	1990 Version 2.05
	1990 Version 2.053
	1990 Version 2.073
	1990 Version 2.083

<table>
<tr>
<td>

```
                    D E R I V E
              A Mathematical Assistant

                    Version 2.50


          Copyright (C) 1988 through 1992 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

If you have received this product as "shareware" or "freeware", you have an
unauthorized copy, because it is a violation of our copyright to distribute
DERIVE on a free trial basis.

To obtain a licensed copy, or if you know of any person or company distributing
DERIVE as shareware or freeware, please contact us at the above address or fax
(808) 735-1105.

                    Press H for help

OPTIONS DISPLAY: Mode: Text Graphics  Reso: Medium(High) Set: Standard(Extended)
                 Adapter: MDA(CGA)EGA MCGA VGA Hercules AT&T T3100 PCjr
Select display mode
                                   Free:100%              Derive Algebra
```

</td>
<td>

1992

Version 2.50

</td>
</tr>
<tr>
<td>

```
                    D E R I V E
              A Mathematical Assistant

                    Version 2.55


          Copyright (C) 1988 through 1992 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

If you have received this product as "shareware" or "freeware", you have an
unauthorized copy, because it is a violation of our copyright to distribute
DERIVE on a free trial basis.

To obtain a licensed copy, or if you know of any person or company distributing
DERIVE as shareware or freeware, please contact us at the above address or fax
(808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA(CGA)EGA MCGA VGA Hercules AT&T T3100 PCjr Other
Select display mode
                                   Free:100%              Derive Algebra
```

</td>
<td>

1992

Version 2.55

The new Adapter “Other” allowed specifying a numerical value for text and graphics modes. This option required BIOS INT 10h support for accessing the graphics screen.

</td>
</tr>
<tr>
<td>

```
                  D E R I V E   X M
              A Mathematical Assistant

                    Version 2.55


          Copyright (C) 1988 through 1992 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

If you have received this product as "shareware" or "freeware", you have an
unauthorized copy, because it is a violation of our copyright to distribute
DERIVE on a free trial basis.

To obtain a licensed copy, or if you know of any person or company distributing
DERIVE as shareware or freeware, please contact us at the above address or fax
(808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA(CGA)EGA MCGA VGA Hercules AT&T T3100 PCjr Other
Select display mode                                          Derive XM
                                   Free:100%                   Algebra
```

</td>
<td>

1992

Version 2.55 XM

The 32-bit DOS extender used by this version required an 80386 to access the larger memory space.
The help menu lists a memory size of 15’097’856 bytes, whereas the 16-bit versions show 621’920 bytes.

</td>
</tr>
<tr>
<td>

```
                  D E R I V E   X M
              A Mathematical Assistant

                    Version 2.57


          Copyright (C) 1988 through 1993 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

If you have received this product as "shareware" or "freeware", you have an
unauthorized copy, because it is a violation of our copyright to distribute
DERIVE on a free trial basis.

To obtain a licensed copy, or if you know of any person or company distributing
DERIVE as shareware or freeware, please contact us at the above address or fax
(808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 PCjr Other
Select display mode                                          Derive XM
                                   Free:100% Ins               Algebra
```

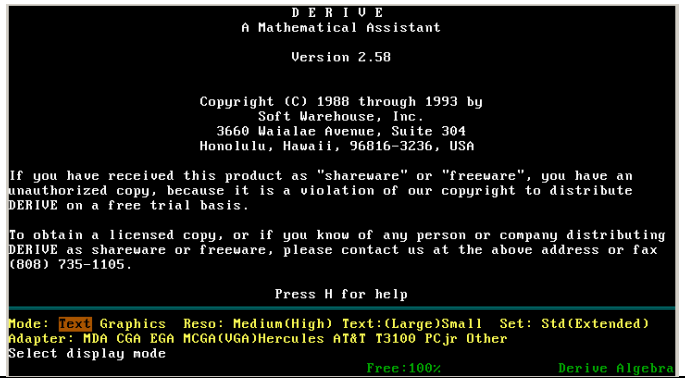
</td>
<td>

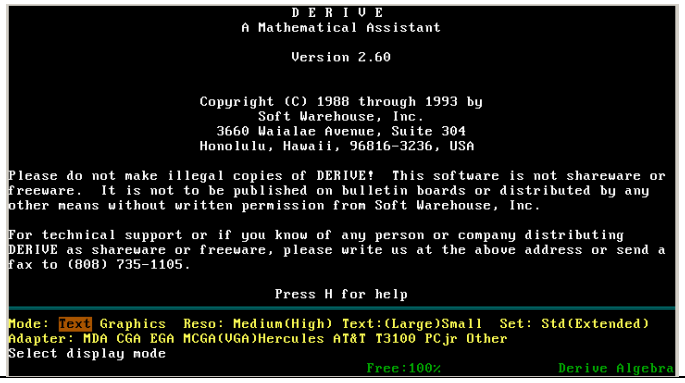
1993

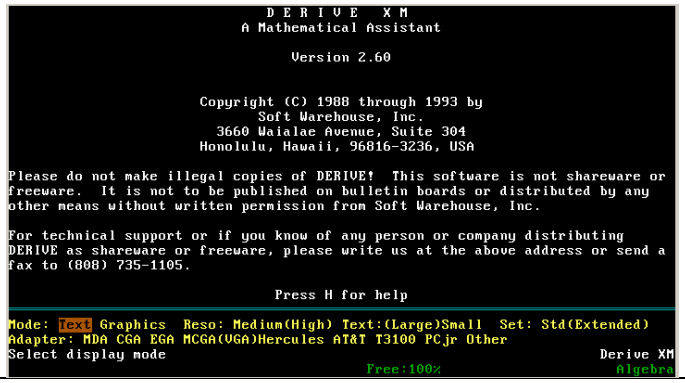
Version 2.57 XM

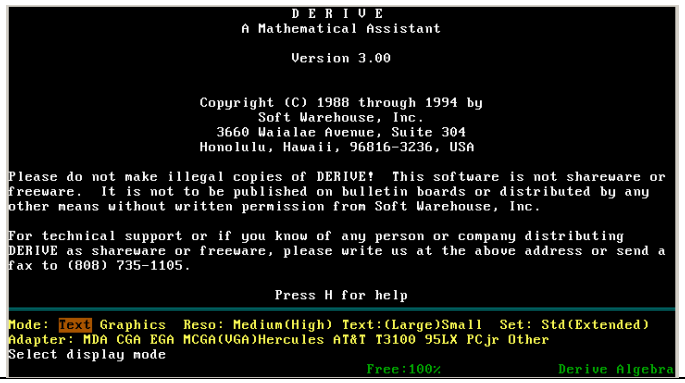
Again a variant using the 32-bit DOS extender.

</td>
</tr>
</table>

<table>
<tr>
<td>


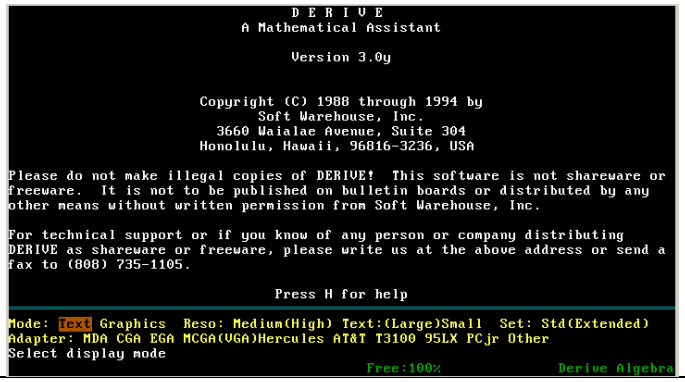
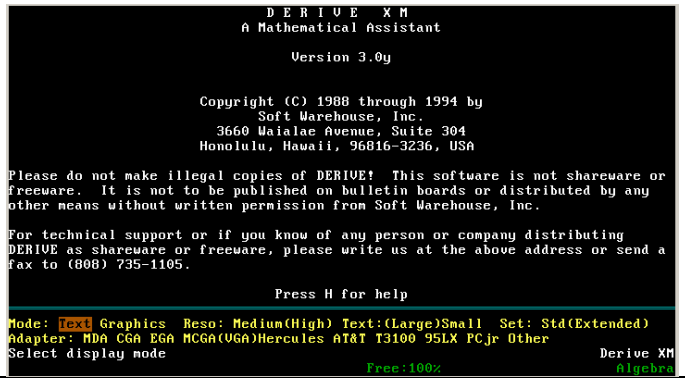
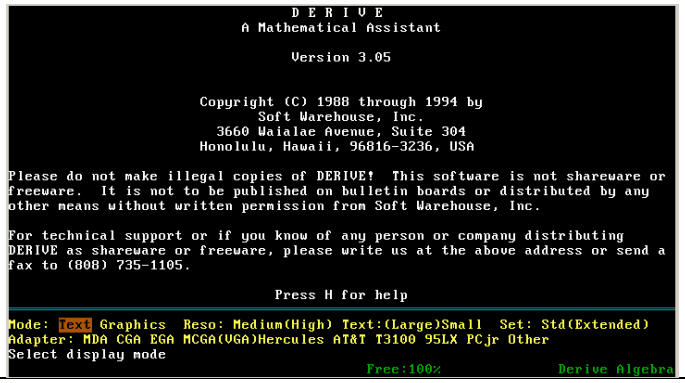
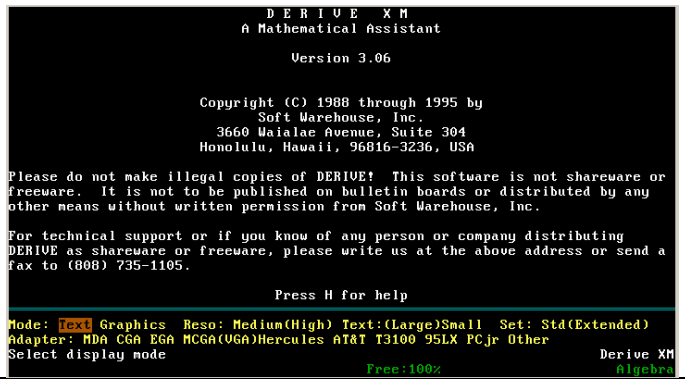
</td>
<td>1993
Version 2.58</td>
</tr>
<tr>
<td>


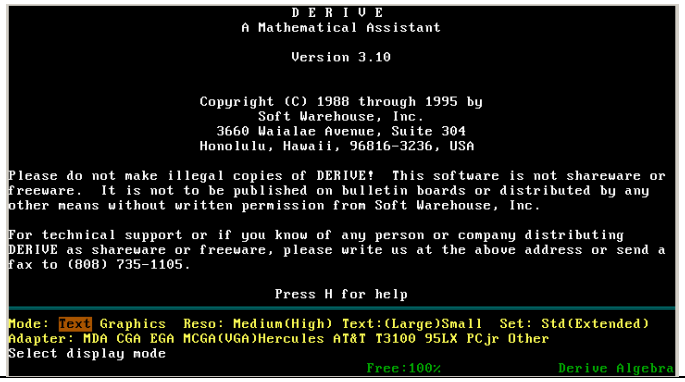
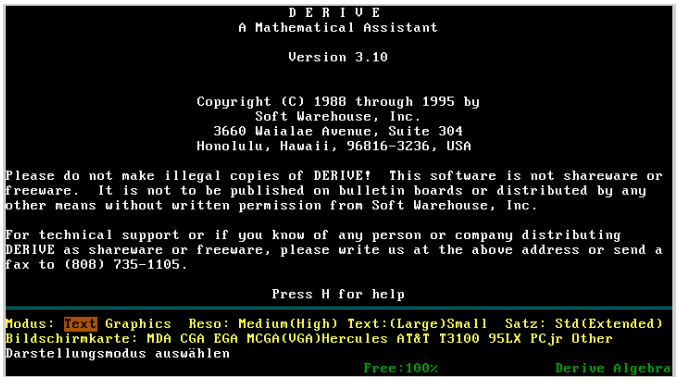
</td>
<td>1993
Version 2.60</td>
</tr>
<tr>
<td>


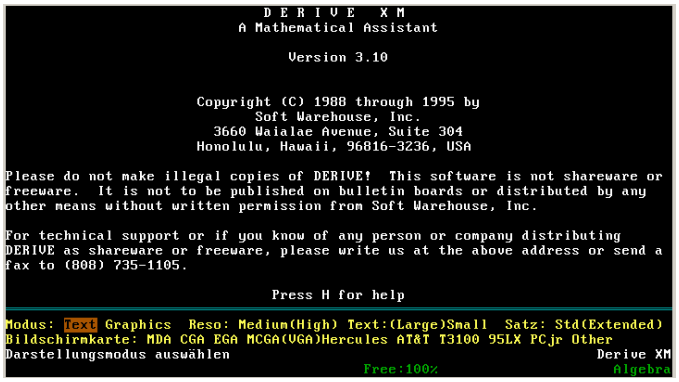
</td>
<td>1993
Version 2.60 XM</td>
</tr>
<tr>
<td>


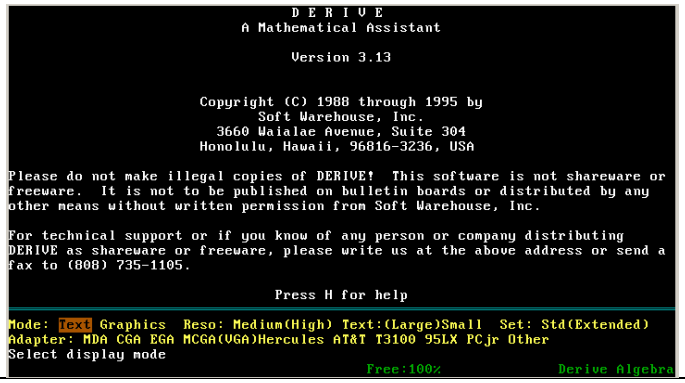
</td>
<td>1994
Version 3.00

This version added support for the display of the HP 95LX palmtop computer.</td>
</tr>
</table>

	1994 Version 3.0y
	1994 Version 3.0y XM This version used a 32-bit DOS extender for accessing up to 15097856 bytes of memory.
	1994 Version 3.05
	1995 Version 3.06 XM This version also used the 32-bit DOS Extender.

<table>
<tr>
<td> 
</td>
<td>1995
Version 3.10</td>
</tr>
<tr>
<td> 
</td>
<td>1995
Version 3.10 G

This version has its user interface translated into German (more obvious in the main and sub-menus).
Localized text is provided in the files “DERIVE.LAN” and “DERIVE.MEN”.</td>
</tr>
<tr>
<td> 
</td>
<td>1995
Version 3.10 XMG

This version used the DOS extender and also came with the user interface translated into German.</td>
</tr>
<tr>
<td> 
</td>
<td>1995
Version 3.13</td>
</tr>
</table>

<table>
<tr>
<td>

```
                    D E R I V E
              A Mathematical Assistant

                   Version 3.14


          Copyright (C) 1988 through 1995 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

</td>
<td>

1995

Version 3.14

</td>
</tr>
<tr>
<td>

```
                 DERIVE for DOS
              A Mathematical Assistant
                   Version 4.04
             running in 16-bit mode


          Copyright (C) 1988 through 1996 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

```
                 DERIVE for DOS
              A Mathematical Assistant
                   Version 4.04
             running in 32-bit mode


          Copyright (C) 1988 through 1996 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

</td>
<td>

1996

Version 4.04

This version came with executables for a 16-bit as well as for a 32-bit DOS Extender variant.

</td>
</tr>
<tr>
<td>

```
                 DERIVE for DOS
              A Mathematical Assistant
                   Version 4.05
             running in 16-bit mode


          Copyright (C) 1988 through 1996 by
                Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

</td>
<td>

1996

Version 4.05

This version came with executables for a 16-bit and a 32-bit DOS Extender variant.

</td>
</tr>
</table>

<table>
<tr>
<td>

```
                    DERIVE for DOS
              A Mathematical Assistant
                    Version 4.05
                 running in 32-bit mode


          Copyright (C) 1988 through 1996 by
                 Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

</td>
<td></td>
</tr>
<tr>
<td>

```
                    DERIVE for DOS
              A Mathematical Assistant
                    Version 4.06
                 running in 16-bit mode


          Copyright (C) 1988 through 1996 by
                 Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

```
                    DERIVE for DOS
              A Mathematical Assistant
                    Version 4.06
                 running in 32-bit mode


          Copyright (C) 1988 through 1996 by
                 Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

</td>
<td>

1996

Version 4.06

This version also provided a variant using the 32-bit DOS Extender.

</td>
</tr>
<tr>
<td>

```
                    DERIVE for DOS
              A Mathematical Assistant
                    Version 4.11
                 running in 16-bit mode


          Copyright (C) 1988 through 1996 by
                 Soft Warehouse, Inc.
            3660 Waialae Avenue, Suite 304
          Honolulu, Hawaii, 96816-3236, USA

Please do not make illegal copies of DERIVE!  This software is not shareware or
freeware.  It is not to be published on bulletin boards or distributed by any
other means without written permission from Soft Warehouse, Inc.

For technical support or if you know of any person or company distributing
DERIVE as shareware or freeware, please write us at the above address or send a
fax to (808) 735-1105.

                    Press H for help

Mode: Text Graphics  Reso: Medium(High) Text:(Large)Small  Set: Std(Extended)
Adapter: MDA CGA EGA MCGA(VGA)Hercules AT&T T3100 95LX PCjr Other
Select display mode
                                   Free:100%            Derive Algebra
```

</td>
<td>

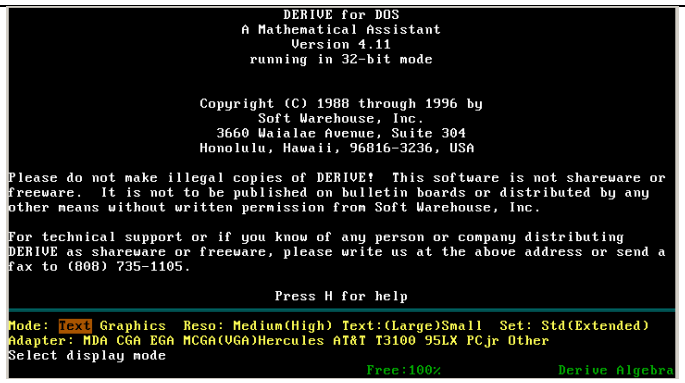
1996

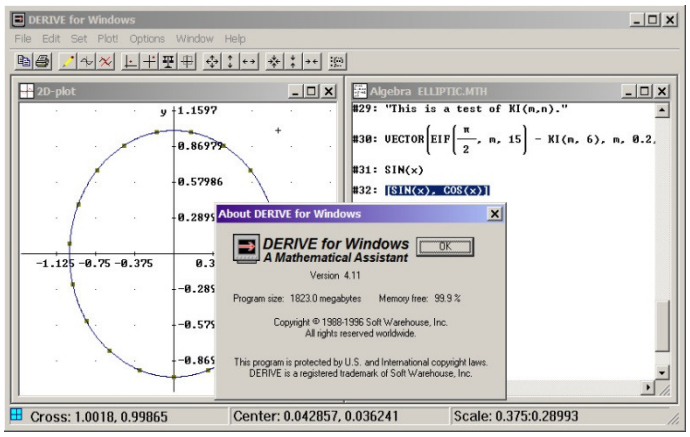
Version 4.11

This version also came with the 32-bit DOS Extender variant.

It still supported the HP 95LX palmtop computer.

</td>
</tr>
</table>





1996

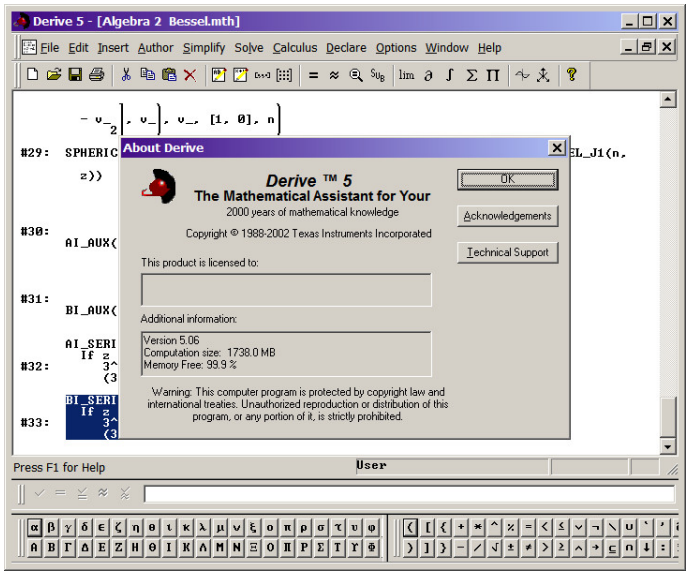
Version 4.11

Windows

This version did not yet have the edit bar at the bottom but used a dialog for authoring equations.

Soft Warehouse Copyright Notice.

It still worked under Windows 7 and 11.



2002

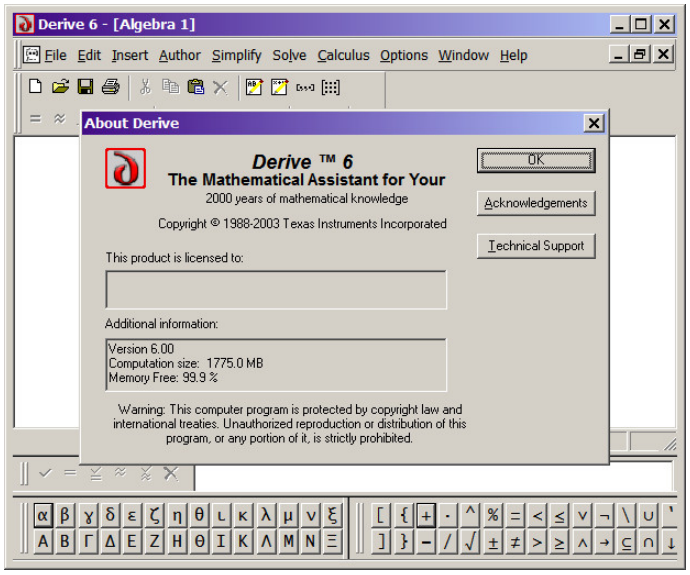
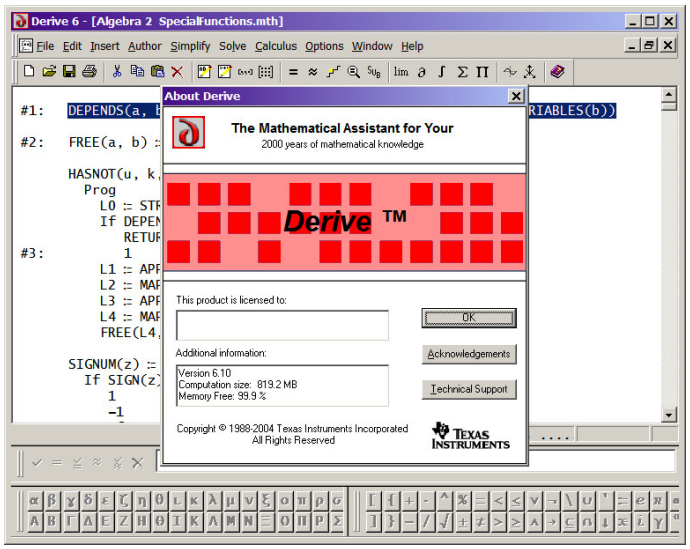
Version 5.06

Windows

Here we see the symbol palettes and the edit bar at the bottom.

Texas Instruments Copyright Notice.

It still worked under Windows 7 and 11.

	2003 Version 6.00 Windows Texas Instruments Copyright Notice.
	2004 Version 6.10 Windows Texas Instruments Copyright Notice. It still worked under Windows 7 and 11.

Terminal Installation for muLISP and Derive

The early versions of muLISP up to version 6.10 and Derive up to version 2.083 tried to find out on which system they were started and chose the appropriate terminal driver. In case the auto-detection scheme failed, they presented a screen with the available terminal types. It is also possible to patch the executable file for a specific terminal.

The terminal list also includes the HP-110 and HP-150. Therefore, Derive versions up to 2.083, which have been built with these muLISP versions, can be patched to work with the HP-150 and HP-110. However, graphics mode and equation displays are not properly supported (axes and points use incorrect characters and would require additional patches).

The *.COM files contain a list of terminals.

```
1 = Other generic MS-DOS computer
2 = IBM PC compatible computer
3 = ANSI.SYS screen or VT-100 Terminal
4 = TI Professional Computer
5 = Zenith Z-100 Computer or VT-52 Terminal
6 = Hewlett-Packard HP-150 Computer
7 = Hewlett-Packard HP-110 Computer
8 = NEC Advanced PC or ADM-3A Terminal
9 = NEC PC-9801 Computer
A = Fujitsu Computer
```

The terminal byte or word in the following context contains the terminal number. Here, it has been changed to “06”, the HP-150 Computer.

```
00000440 00001E1E 04000300 00004F00 0000FF07 ..........O.....
00000450 5C7C64FF FF1A0006 00000000 00242000 \|d..........$ .
                        ^^ terminal type
00000460 0000004C 5350FF00 002E1110 FF000000 ...LSP..........
```

muMATH

Before Derive there was muMATH. It was written in the muSIMP language which itself was implemented in muLISP. muSIMP provided a simpler and more conventional user interface to muLISP.

Due to the memory restrictions of the contemporary systems (initially 8080 systems running CP/M, CDOS, IMDOS, TRSDOS), a hierarchical subset of the available files has to be loaded, depending on the problem to solve. The system of user-selected components could then be saved to reduce the startup time.

The figure below shows the dependency tree of these packages. For example the approximation of a function by a Taylor series requires reading the sequence of files ARITH.MUS, ALGEBRA.ARI, DIF.ALG and, TAYLOR.DIF. Note that the file extensions indicate the required package, starting with MUS for the muSIMP core.

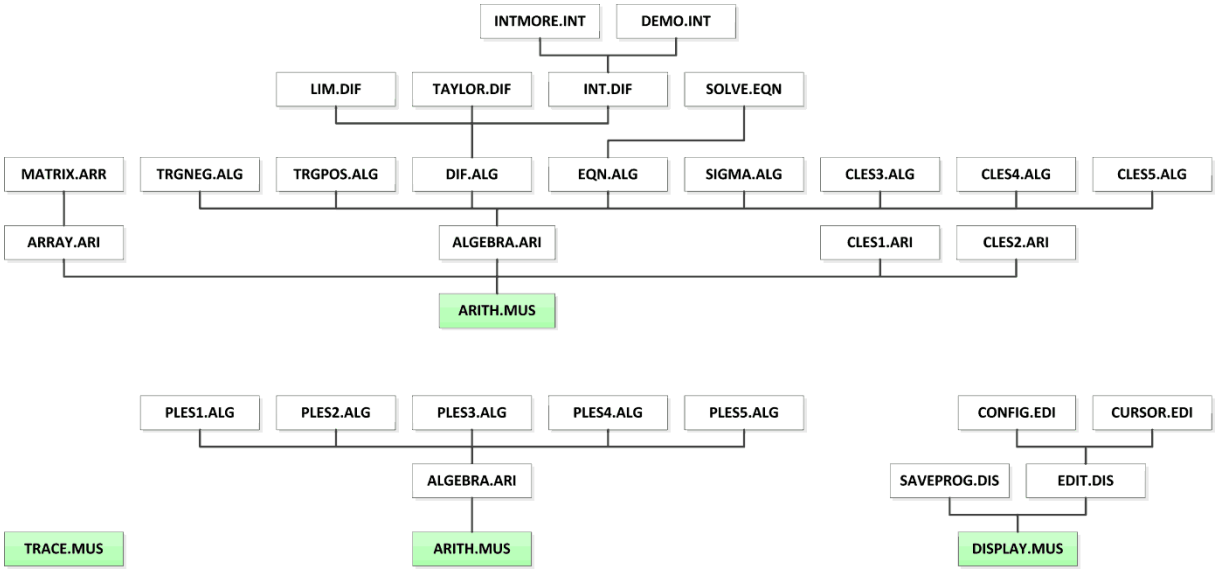


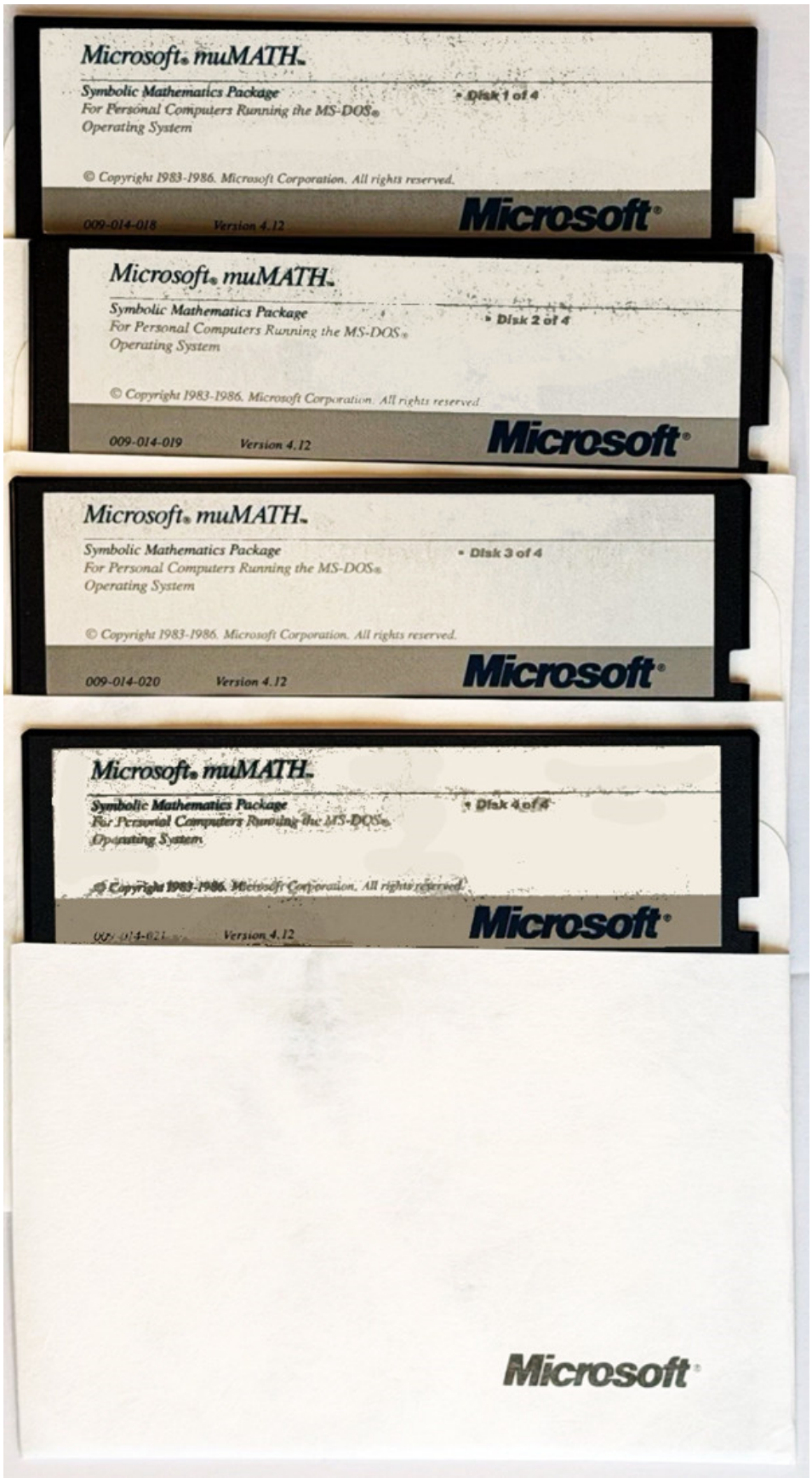
Figure 5: Dependency tree of muMATH 2.15 for CP/M modules. Start loading at one of the green nodes, following the tree up to the final module. Versions for MS-DOS included additional modules and preloaded combinations in SYSTEM files.

Following the generic 8080 CP/M version, native TRS-80 and Apple II versions and at least two versions for MS-DOS (muMATH-82 and muMATH-83) have been developed.

The TRS-80 version of muMATH-79 also provides access to a few low level TRS-80 graphics functions like CLS(), POINT(x,y), SET(x,y) and, RESET(x,y).

Many of these very early versions seem to be missing from the known internet archives. For example, the initial versions muMATH-79 for CP/M as well as muMATH-82 for MS-DOS seem to have been lost.

The last version for MS-DOS was muMath-83 Version 4.12, delivered by Microsoft on four 5-¼ disks. Here, the two core files MUSIMP.COM and MUMATH.SYS have been preserved, but additional files and manuals are missing.



A set of muMath 4.12 disks for MS-DOS, lifted from EBay in 2026.

```
muSIMP-80  2.02  COPYRIGHT (C) 1980 MICROSOFT
LICENSED FROM THE SOFT WAREHOUSE

? RDS(ARITH,MUS,A);

@: ARITH


? RDS(ALGEBRA,ARI,A);

@: ALGEBRA


? RDS(DIF,ALG,A);

@: DIF


? RDS(TAYLOR,DIF,A);

@: TAYLOR


? TAYLOR(SIN(X),X,0,2);

@: X*COS(0) - X^2*SIN(0)/2 + SIN(0)
```

muMATH 2.02

8080

CP/M Version

1980

This version was for generic CP/M systems using the 8080 and Z80 processors.

```
muSIMP-80 2.12 (07/09/81)
Copyright (C) 1981 MICROSOFT
Licensed from The SOFT WAREHOUSE

? RDS(ARITH,MUS);
@: ARITH


? RDS(ALGEBRA,ARI);
@: ALGEBRA


? RDS(DIF,ALG);
@: DIF


? RDS(TAYLOR,DIF);
@: TAYLOR


? TAYLOR(SIN(X),X,0,2);
@: X*COS(0) - X^2*SIN(0)/2 + SIN(0)
```

muMATH 2.12

8080

CP/M Version

1981

Besides the version number, there is no visible difference to version 2.02.

```
APPLE ][ ADIOS-81          VERSION 01/29/82

 20 MUSIMP      COM  ! 20 MUSIMPX     COM
  4 TRACE       MUS  ! 17 ARITH       MUS
 11 ALGEBRA     ARI  !  2 EQN         ALG
  4 SOLVE       EQN  !  5 ARRAY       ARI
  7 MATRIX      ARR  !  2 LOG         ALG
  3 TRGPOS      ALG  !  4 TRGNEG      ALG
  3 DIF         ALG  !  6 INT         DIF
  8 INTMORE     INT  !  1 TAYLOR      DIF
  9 LIM         DIF  !  4 SIGMA       ALG
  1 DISKCOPY    COM  !

FREE ON 1:    1 K-BYTES

  7 CLES1       ARI  !  6 CLES2       ARI
 15 CLES3       ALG  ! 10 CLES4       ALG
 12 CLES5       ALG  ! 19 PLES1       TRA
  9 PLES2       TRA  ! 14 PLES3       TRA
 13 PLES4       TRA  ! 14 PLES5       TRA
  1 DISKCOPY    COM  !  5 DEMO        ALL

FREE ON 1:    7 K-BYTES

  7 DISPLAY     MUS  ! 29 EDIT        DIS
  4 SAVEPROG    DIS  !  2 CURSOR      EDI
  1 CONFIG      EDI  !  1 DISKCOPY    COM

FREE ON 2:   88 K-BYTES
```

muMATH with ADIOS was supplied on 3 disks

muMATH 2.15

6502

Apple II

ADIOS Version

1982

This version came with its own ADIOS operating system to maximize the memory available for muMATH.

The native code of muMATH was probably a semi-automatic translation of the 8080 code to the 6502 and not very fast.

Just reading and interpreting the files “ARITH.MUS”,

<table>
<tr>
<td>

```
MUSIMP-80 2.15 (03/01/82)
APPLE ][ ADIOS VERSION
COPYRIGHT (C) 1981 THE SOFT WAREHOUSE
LICENSED BY MICROSOFT, INC.

? RDS(ARITH,MUS,1);
@: ARITH

? RDS(ALGEBRA,ARI,1);
@: ALGEBRA

? RDS(DIF,ALG,1);
@: DIF

? RDS(TAYLOR,DIF,1);
@: TAYLOR

? TAYLOR(SIN(X),X,0,2);
@: X*COS(0) - X^2*SIN(0)/2 + SIN(0)

? ※
```

</td>
<td>

“ALGEBRA.ARI”, “DIF.ALG” and “TAYLOR.DIF”, for the Taylor series expansion, takes almost 10 minutes.

</td>
</tr>
<tr>
<td>

```
FILE DIRECTORY --- DRIVE 0     MUMATH   -- 00/00/00

TAYLOR/INT          MATH48/CMD          LOG/ALG
TRGNEG/ALG          TRGPOS/ALG          DEMO/INT

TRSMATH-80  1.0
COPYRIGHT (C) 1980 MICROSOFT
LICENSED FROM THE SOFT WAREHOUSE

? RDS(TAYLOR,INT,1);

@ TAYLOR

? TAYLOR(SIN(X),X,0,2);

@ X*COS(0) - X↑2*SIN(0)/2 + SIN(0)

? _
```

</td>
<td>

TRSMATH 1.0

TRS-80 Model I

1980

This version of muMATH was provided on two disks, one for 32 and the second one for 48 Kbytes of memory. The 32 KB version omitted partial derivatives and indefinite integration functions. The available disk image contains the 48 KB version.

In this version, TAYLOR.INT must be read to make the Tylor series expansion available.

</td>
</tr>
<tr>
<td>

```
D:\DERIVE\muMATH-83_4.12-MSDOS>musimp MUMATH

muSIMP-83 4.12 (03/30/85)
MS-DOS Version
Copyright (C) 1982 The SOFT WAREHOUSE
Licensed by MICROSOFT Corp.


?  TAYLOR(SIN(X),X,0,2);
@: X

?  _
```

</td>
<td>

muMATH-83 4.12

MS-DOS

1985

This version of muMATH was provided on four disks and provided the functionality of the previous versions without the need for loading additional files. Note that the output of the TAYLOR function has the COS(0) and SIN(0) terms evaluated for improved output.

</td>
</tr>
</table>

Literature related to Derive

There are many books about Derive and its application and quite a few of them have been listed in the Derive Users Group Newsletters. Some can be loaned from your local library or from archive.org. Therefore, the following list contains only a few examples.

[1] Gilligan, Lawrence G., Marquardt, James F. Sr., “Calculus and the Derive Program: Experiments with the Computer”, Gilmar Publishing, 1991, 152 pages.

[2] Arney, David C., “Derive Laboratory Manual for Differential Equations”, Addison-Wesley, 1991, 189 pages.

[3] Arney, David C., “Exploring Calculus with Derive", Addison-Wesley, 1992, 166 pages.

[4] Arney, David C., “The Student Edition of Derive”, Addison-Wesley, 1992, 387 pages, uses Derive Version 2.

[5] various authors, “Lab Resource Manual to accompany The Student edition of Derive”, 1992, 69 pages.

[6] Glynn, Jerry, “Exploring Math from Algebra to Calculus with Derive”, Mathware, 1992, 154 pages, uses Derive Version 2.51.

[7] Simon, B., “It’s Not Just for Mainframes Anymore”, PC Magazine, August 1992, pp.405-433. [software review, including Derive 2.1]

[8] Berry, J.S., Graham, E., Watkins, A. J. P., “Learning Mathematics through Derive”, Ellis Horwood, 1993, 371 pages.

[9] Koepf, W., Ben-Israel, Ben, Gilbert, Robert P., “Mathematik mit Derive”, Vieweg 1993. [German]

[10] Denton, Brian, “Learning Linear Algebra through Derive”, Prentice Hall, 1995, 353 pages.

[11] Townend, M. Stewart, Pountney, David C., “Learning Modelling with Derive”, Prentice Hall, 1995, 244 pages.

[12] Richardson, R. L., “Business Calculus today with Spreadsheets and DERIVE”, Saunders College Publishing, 1996, 416 pages.

[13] Abbey, May Kay, “Calculus Explorations using Derive”, Saunders College Publishing, 1996, 84 pages.

[14] Bogess, Al, et. al., “Single Variable Calculus with Derive”, Brooks/Coole Publishing Company, 1999, 208 pages.

[15] Roanes-Lozano, E., Galán-García, J. L., Solano-Macías, C., “Some Reflections About the Success and Impact of the Computer Algebra System DERIVE with a 10-Year Time Perspective”, Mathematics in Computer Science, 2019, pp. 417-431.

Literature related to muMATH and muLISP

[16] Rich, A.D., Stoutemyer, D. R., “Capabilities of the muMATH-79 Computer Algebra System for the Intel-8080 Microprocessor”, in “Symbolic and Algebraic Computation, EUROSAM 1979”. Lecture Notes in Computer Science, Vol 72. Springer, 1979.

[17] Williams, G., “The muSIMP/muMATH-79 Symbolic Math System”, BYTE Magazine 11, 1980, pp. 324-338.

[18] Robinson, T., “Honolulu's innovatory packages”, Practical Computing, November 1980, pp.60-61. [muLISP-79, muSIMP-79, muMATH-79]

[19] Wyant, James C., "Use of a Symbolic Math System to Solve Polarized Light Problems", Applied Optics, Vol. 20, No. 19, 1 October 1981, pp.3321-3326. [application of muMATH-79]
[20] Levisky, J., “muLISP: Latest in a Long List of LISPs”, InfoWorld, V3N15, 27 July, 1981, pp. 10-13. [muLISP-80]
[21] Shochat, David D., “Experience with the muSIMP/muMATH-80 Symbolic Mathematics System”, ACM SIGSAM Bulletin #3, August 1, 1982, pp. 16-23. [refers to muSIMP/muMATH-79]
[22] Halamka, J., “MuLISP/MuStar-80 develops, edits, debugs LISP”, InfoWorld, V5N10, 7 March, 1983, pp. 54-55. [muLISP-80]
[23] McClennan, David T., “LISPing with your PC”, (review, includes muLISP-82), PC Magazine, December, 1983.
[24] Wong, William G., “The PC Speaks LISP”, (review, includes muLISP-82), PC Tech Journal, April, 1984, pp. 112-148.
[25] Bortz, J., Diamant J., “LISP for the IBM Personal Computer”, (review, includes muLISP-83), BYTE Magazine, July, 1984.
[26] Warusfel, A., Marx, J.-L., “Mu-math met de l’algèbre dans votre machine”, L’Ordinateur Individuel, No. 63, Octobre 1984, pp. 134-137.
[27] Barcellos, A., “Infinite Precision”, PC-Magazine December 11, 1984, pp. 379-380.
[28] Stoutemyer, D.R., “A preview of the next IBM-PC version of muMATH”, in “EUROCAL '85”, Lecture Notes in Computer Science, Vol 203, Springer, 1985.
[29] Wong, William G., “16 Bit Lisp and Prolog Implementations”, (review, includes muLISP-82), Micro/Systems Journal, Part I: Vol. 01, No. 01, March/April, Part II: Vol. 01, No. 02, May/June, 1985.
[30] Rosenbeck, P., Rainer, J., “Lisp für Mikros”, (review, includes muLISP 4.1), c’t Magazin, 3, 1986. [German]
[31] Piddock, P., “Extended muSIMP/muMATH for Teaching and Learning Mathematics”, Comput. Educ., Vol. 10, No. 1, Pergamon Press, 1986, pp. 155-158.
[32] Schwartz, Stanley, “Customizing muLISP”, Sextant, Issue 20, Jan-Feb 1986. [refers to muLISP-83 CP/M, muLISP-85 MS-DOS and Zenith computers]
[33] Trindle, Carl, “Application of the MuMATH Symbol Manipulation System to Chemically Significant Permutation Groups”, J. Symbolic Computation, 1986, pp. 207-212. [refers to Apple II 6502 version of muMATH]
[34] Wooff, C., Hodgkinson, D., “muMath - a Microcomputer Algebra System”, Academic Press, 159 pages, 1987.
[35] Reid, C., “muMATH Program”, PC-Magazine, April 14, 1987, pp. 173-174 [Review of muMATH 4.12]
[36] DeMers, Michael N., “SEDRULE: A Rule-Based System for Interpreting some Major Sedimentary Environments”, Computers & Geosciences, Vol. 16, No. 6, 1990, pp.833-846. [application of muLISP-86]
[37] Handke, J., “Sprachverarbeitung mit LISP und PROLOG auf dem PC”, Vieweg, Programmieren von Microcomputern 27, 1987. [German, application of muLISP-86]
[38] Deiß, K., Handke, J., Meyer, B., “Professionelles Programmieren mit LISP”, McGraw-Hill, 606 pages, 1990. [German, includes an excellent description and many examples for integrating assembly language modules into muLISP-86]

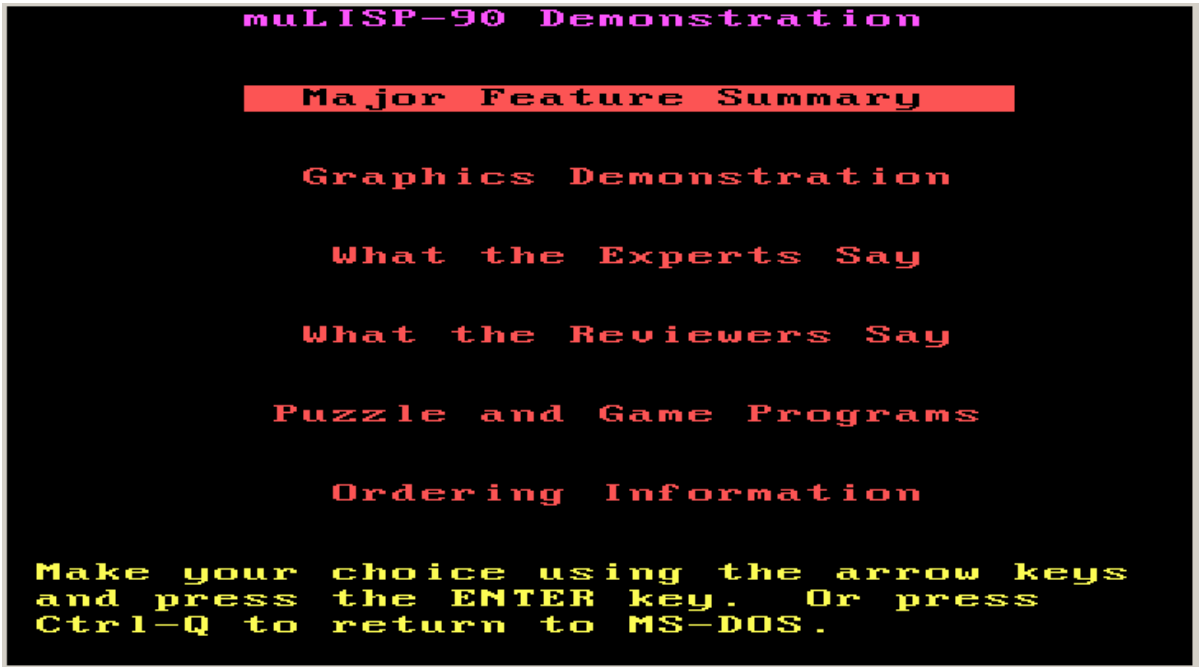
[39] Schoffa, G., “Professionelle Lisp-Interpreter für PC's”, mc Magazin, Februar 1988, p. 126-128, Franzis Verlag. [German, brief comparison of muLISP-87 an CGLisp 1.02]

Software Manuals

[40] “muSIMP/muMATH-79 Reference Manual”, 135 pages, 1979.
[41] “muLISP/muSTAR-80 Artifical Intelligence Development System”, Reference Manual, The Soft Warehouse, 1980.
[42] “muMATH/muSIMP”, for TRS-80, software manual, Microsoft, 76 pages, 1980.
[43] “The muMATH/muSIMP-80 Symbolic Mathematics System”, Reference Manual, 195 pages, 1980. [covers TRS-80, Cromenco Z1&Z2&Z3, Imsai VDP IMDOS, Apple II CP/M with Z80 card, Heath H89 with CP/M board, 8080, 8085, Z80 CP/M systems]
[44] “The muMATH/muSIMP-80 Symbolic Mathematics System for the Apple II with SoftCard”, Reference Manual, 210 pages, 1981. [covers Apple II CP/M with Z80 card]
[45] “The muMATH/muSIMP-80 Symbolic Mathematics System for the CP/M Version”, Reference Manual for the CP/M Version, 148 pages, 1981. [covers Apple II CP/M with Z80 card, Apple II with Apple DOS 3.3, TRS-80 1&3 TRSDOS, TRS-80 2 CP/M, Imsai VDP IMDOS, 8080, 8085, Z80 CP/M systems]
[46] “Microsoft muLISP Artificial Intelligence Development System”, Reference Manual for muLISP-83, 1983. [covers versions for CP/M, IBM PC, Apple II with Z80 card]
[47] “muLISP-87 Reference Manual”, 2nd printing, Soft Warehouse Inc., 490 pages, January 1988.
[48] “muLISP-90 LISP Language Programming Environment”, Reference Manual, 1990.

Software Extensions

[49] Schwartz, S., “muLISP-87 Connection”, Z-100 LifeLine, Public Domain and ShareWare Library, #176, 1988. [a Z-100 interface library for muLISP-87]
[50] Edgar, G. A., “muMATH 2.12 Enhancements”, CP/M Users Group, Vol. 83, March 1, 1982.
[51] Carter, M., “Adding I/O Functions to muLISP”, Dr. Dobbs Journal, Vol. 9, 1984. [refers to muLISP-80 CP/M]



The demonstration program for muLISP-90 consisted of a linked set of colorful text screens in presented in CGA mode.